Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think I'm going to have to do a lot of lists and loops. I hope to improve on loops

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think lists because im pretty sure thats what im mostly gonna use to make the profiles.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Mostly I think the dataclass would be the first thing I do.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I think getting tired or other classes will get in the way of my work. But my goal of getting extra help and going to the library could help.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

roomate matcher

1. Create project

Tell us about yourself

First Middle last

hobies major Clean

early bird sleep in

result

Jake L	95% match	reason why
Tony T	90% match	reason why
Isat N	80% Match	reason why
Luke J	70% Match	reason why

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	Helps college students find a compatible roommate. The User enters their information, and it will find their best match.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	**Profile: list[profile]** all student profiles, starting with some sample ones A new profile is .append in **save_profile** **Current_name:str** the name of the users who just made the profile. Its set in **save_profile** and used to find there profile. **Matches: list[match]** the ranked results for the current user, it get recalculated and rebuilt every time a match is calculate. **Message: str** a status or error message(for example: enter your name) **Filter_goal:str** roommate searching or study partner searching.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified	Profile:str Name:str Sleep_schedule:str Clean:str

(if they are modified)	noise _level:str Bio:str guest:bool
Routes (minimum 7) The names of your route functions	Index, create_profile,save_profile,show_result,show_all_profiles, filter_profiles, how_scoring_work,edit_profile
Pages (minimum 5) A brief description of the pages in your site	Home: short intro and buttons for other pages Create profile: creating a profile page Result: ranked matches All profiles: shows all profiles How scoring works: shows how they find the best match
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Textbox: for name Selectbox: for sleeping schedules, ect check Box: for smokers and guest Text area: for short bios
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	**Save_profile** if the name is empy show a error message instead of saving. But basically itll show you all the boxes you need to answer **Filter_profile** filters all pages and only shows profiles with that goal **Show_result** if theres no matches, show no matches if so show matches

Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	State.profile compares the new profiles, scores each on and builds a match.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Did not use

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.